Technical Explanation SKAI3 LV ® Eval-Kit Adapter Board

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Keyword: eval-kit; adapter board; SKAI3LV; pin-out

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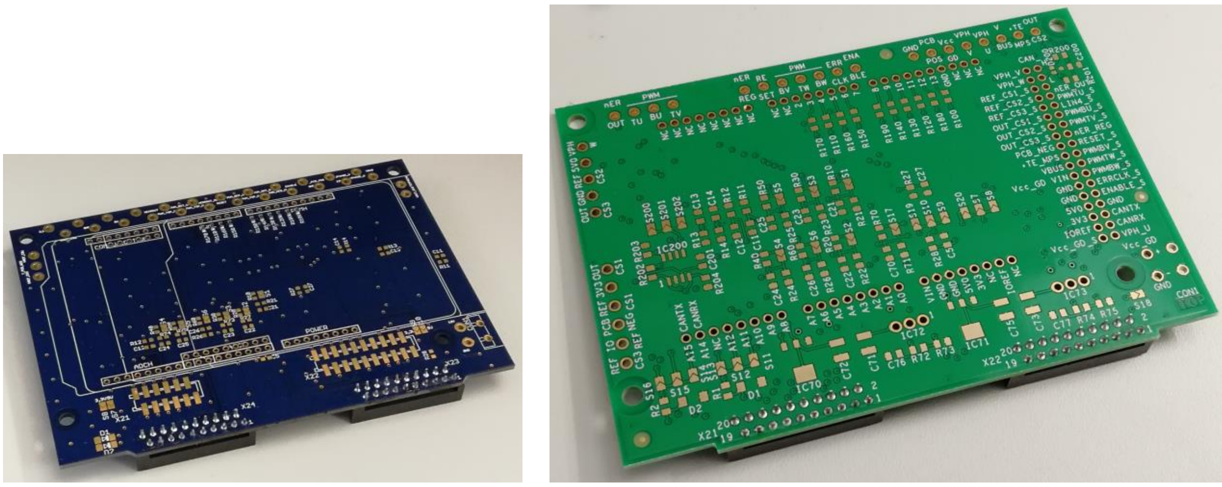
1. Note for Use of this Document

This document describes the SKAI3 LV eval-kit adapter board, which is delivered with every SKAI3 LV Evaluation Version.
The SKAI3 LV eval-kit adapter board allows fast adoption of the SKAI3 LV to already existing motor control systems. It is suited for evaluation and demonstration purposes e.g. before starting a series design.
For description of the SKAI3 LV refer to [3].

This Technical Explanations version describes the **new** adapter board **layout** (see Figure 1, **green** board) with substantial changes to the previous PCB layout.
For description of the **previous** adapter board layout (Figure 1, **blue** board) please refer to the outdated document version [4].

This Technical Explanations document may be subject to changes.

Figure 1: blue: previous adapter board version / green: adapter board described in this document version



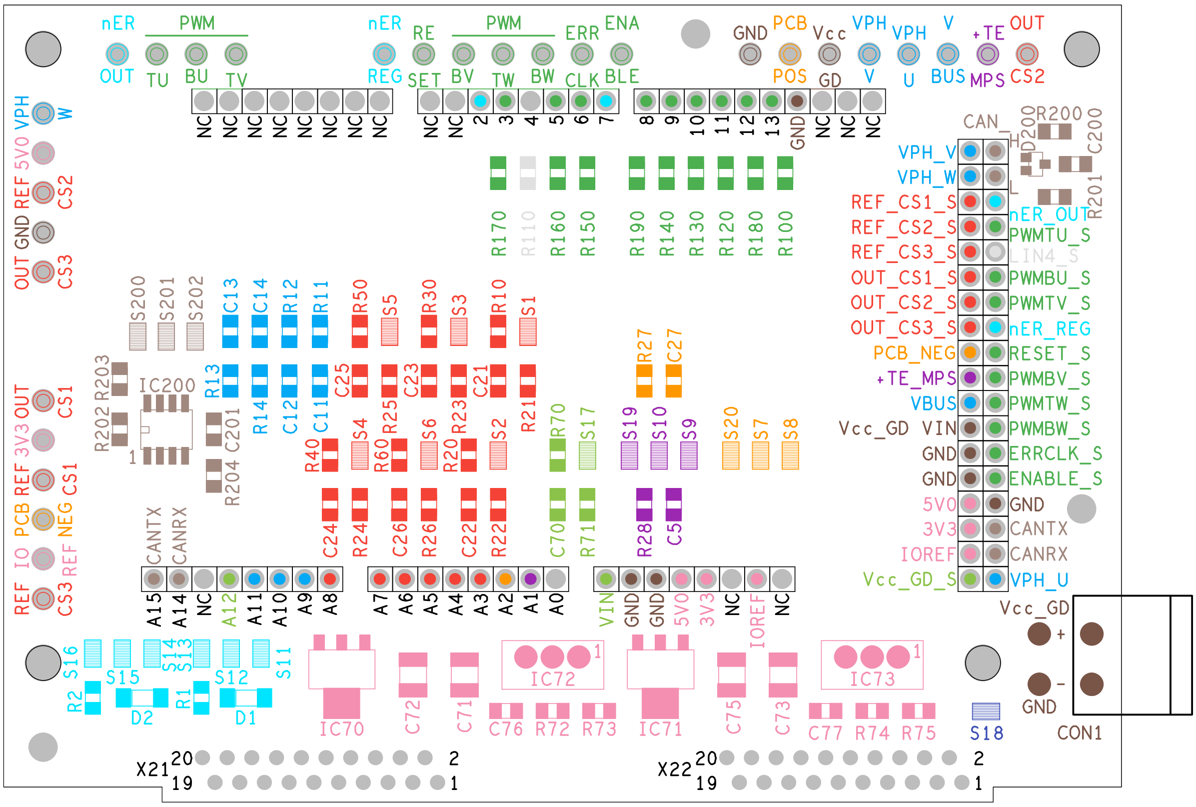
2. Functions and interfaces

2.1 SKAI3 LV controller interface

The eval-kit adapter board connects a customer controller board with the SKAI3 LV across signal conditioning circuits. The signal conditioning circuits are described in the following chapters.

Note: Assemblies of circuit components to meet controller system requirements/signal levels or SEMIKRON recommendations are required before operating the SKAI3 LV.

Figure 2: Eval-kit adapter board assembly (for details of coloured sections refer to subsequent document chapters)

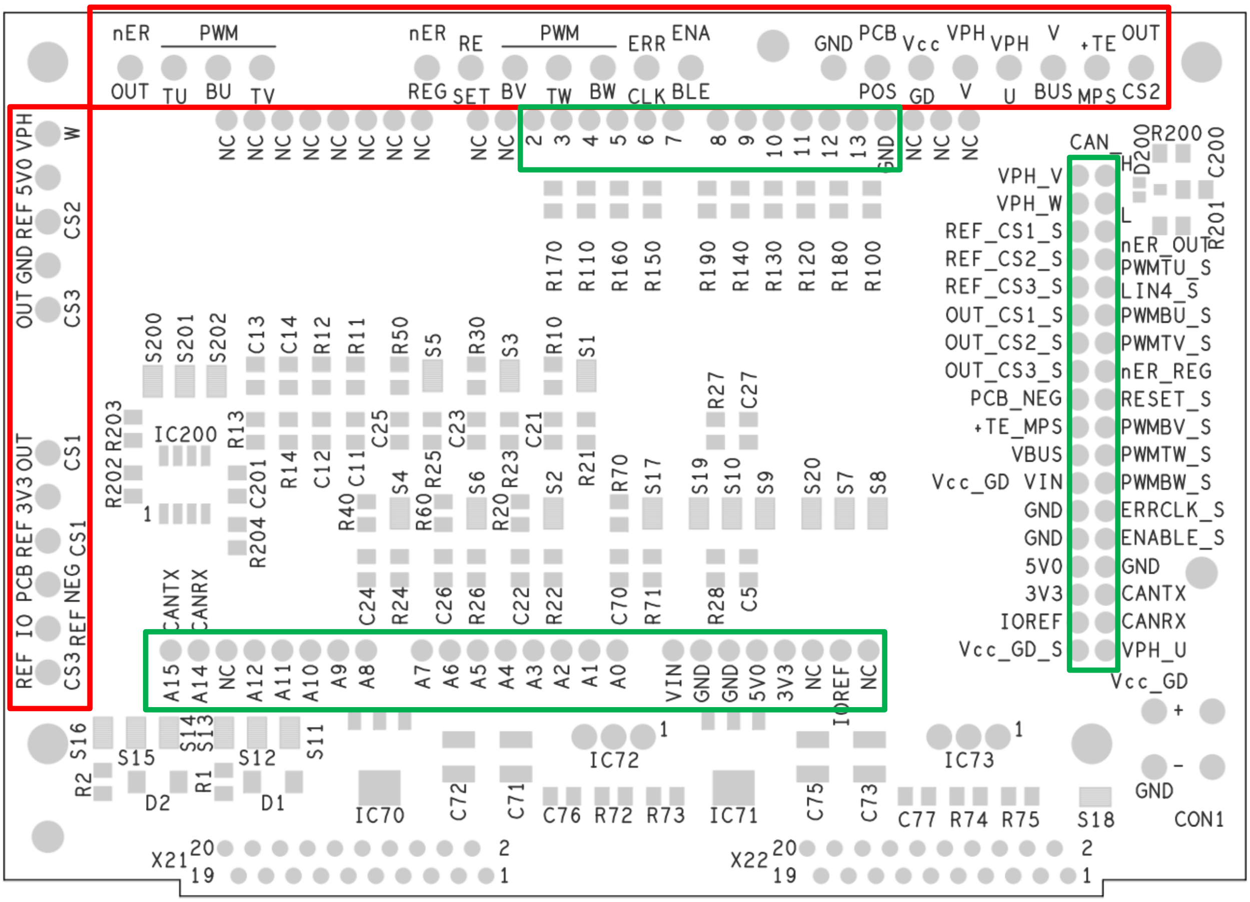


Board supply:
Connector for 13.0 V supply voltage for SKAI3 LV (dark brown)
Optional connection of 13.0 V supply voltage (or scaled value) for use on controller board (light green)

Signal Conditioning:
Digital input signals (green)
Current signals (red)
Phase voltage signals / DC link voltage signal (blue)
DCB temperature signal (purple)
PCB temperature signal (orange)
Pull-up options and protection diodes for digital output signals (turkey)

Additional:
Optional connection between GND and DC minus potential (dark blue)
Optional CAN Transceiver + periphery (light brown)
Optional 5V, 3.3 V linear power supplies (rose)

Figure 3: Eval-kit adapter board signal assignment



Direct access to signals on X21 and X22
Connection interfaces to controller board

Please note: Signal names with suffix "*_S* indicate SKAI3 LV signals with additional circuitry on the adapter board (e.g. voltage dividers, filters or protective components). Signal names without suffix "*_S" are directly routed thru on the adapter board.

Table 1: Eval-kit adapter signal specification

Signal Name	Description	Type	Electrical characteristics	Arduino Uno/Due	SKAI3 LV
GND	Gate driver common ground	power	0V I_{max} = 1A continuous; 2A for 3s	GND	X21:1, X21:17, X22:5, X22:19
Vcc_GD	Gate driver power supply	power	V = 13V ±5% I_{max} = 250mA		X21:4, X21:8
	Connectable to Vcc_GD via solder jumper; see chapter 2.2	power		VIN	
ENABLE	Gate driver enable, high active low: disable PWM, set error signal active high: enable PWM see chapter 2.3.2	digital input	low: 0V … 0.8V high: 2.4V … Vcc_GD R_{IN} = 50 kΩ		X21:2
ENABLE_S	Connectable to ENABLE via resistor R100 see chapter 2.3.2	digital input		1	
VBUS	Scaled DC ink voltage see chapter 2.5	analogue output	0V … $V_{CC,sens,max}$ -> 0V … 5V	A0	X21:3
VPH_W	Scaled phase L3/W voltage see chapter 2.5	analogue output	0V … $V_{CC,sens,max}$ -> 0V … 5V	A9	X21:5

VPH_V	Scaled phase L2/V voltage see chapter 2.5	analogue output	0V … $V_{CC,sens,max}$ -> 0V … 5V	A10	X21:6
VPH_U	Scaled phase L1/U voltage see chapter 2.5	analogue output	0V … $V_{CC,sens,max}$ -> 0V … 5V	A11	X21:7
PWM_BW	Gate driver phase L3/W low side input, high active see chapter 2.3.2	digital input	low: 0V … 0.8V high: 2.4V … Vcc_GD R_{IN} = 50 kΩ		X21:10
PWMBW_S	Connectable to PWM BW via resistor see chapter 2.3.2	digital input		11	
PWM_TW	Gate driver phase L3/W high side input, high active see chapter 2.3.2	digital input	low: 0V … 0.8V high: 2.4V … Vcc_GD R_{IN} = 50 kΩ		X21:11
PWMTW_S	Connectable to PWM TW via resistor see chapter 2.3.2	digital input		10	
PWM_BV	Gate driver phase L2/V lo side input, high active see chapter 2.3.2	digital input	low: 0V … 0.8V high: 2.4V … Vcc_GD R_{IN} = 50 kΩ		X21:12
PWMBV_S	Connectable to PWM BV via resistor see chapter 2.3.2	digital input		9	
PWM_TV	Gate driver phase L2/V high side input, high active see chapter 2.3.2	digital input	low: 0V … 0.8V high: 2.4V … Vcc_GD R_{IN} = 50 kΩ		X21:13
PWMTV_S	Connectable to PWM TV via resistor see chapter 2.3.2	digital input		6	
PWM_BU	Gate driver phase L1/U low side input, high active see chapter 2.3.2	digital input	low: 0V … 0.8V high: 2.4V … Vcc_GD R_{IN} = 50 kΩ		X21:14
PWMBU_S	Connectable to PWM BU via resistor see chapter 2.3.2	digital input		5	
PWM_TU	Gate driver phase L1/U high side input, high active see chapter 2.3.2	digital input	low: 0V … 0.8V high: 2.4V … Vcc_GD R_{IN} = 50 kΩ		X21:15
PWMTU_S	Connectable to PWM TU via resistor see chapter 2.3.2	digital input		3	
ERR_CLK	Clock signal input to read out error register see chapter 2.3.2	digital input	low: 0V … 0.8V high: 2.4V … Vcc_GD Rin = 50 kΩ		X21:16
ERRCLK_S	Connectable to ERR CLK via resistor see chapter 2.3.2	digital input		12	
nER_OUT	Indicates error state, low active Connect a pull-up resistor in range between 2 kΩ to 5.5 kΩ to positive controller I/O supply of respective input pin see chapter 2.3.1	digital output	Open drain; low active; R_{ON}<240 Ω	2	X21:18
RESET	Gate driver reset signal, high active low: disable PWM, set nErrOut signal active high: release PWM rising edge: reset error state	digital input	low: 0V … 0.8V high: 2.4V … Vcc_GD R_{IN} = 50 kΩ		X21:19
RESET_S	Connectable to RESET via resistor see chapter 2.3.2	digital input		8	
nER_REG	Gate driver error register output Connect a pull-up resistor in range between 2 kΩ to 5.5 kΩ to positive controller I/O supply of respective input pin see chapter 2.3.1	digital output	Open drain; R_{ON}<240 Ω	7	X21:20
+TE_MPS	Scaled power module temperature signal referenced to GND	analogue output	see chapter 2.6	A1	X22:6, X22:5 (pulled to GND)

PCB_POS	Positive connection to PCB temperature sensor. Connectable to different voltage levels	analogue output	see chapter 2.7		X21:12
PCB_NEG	Scaled PCB temperature signal referenced to GND	analogue output	see chapter 2.7	A2	X22:11
OUT_CS3	Output signal current sensor phase L3/W	analogue output	$+I_{PM}$ … $-I_{PM}$ -> 0V … 5V see chapter 2.4		X22:13
OUT_CS3_S	Connectable and scalable to OUT_CS3 via resistor or solder jumper	analogue output		A3	
REF_CS3	Reference voltage current sensor phase L3/W	analogue output	V = 2.5V see chapter 2.4		X22:14
REF_CS3_S	Connectable and scalable to REF_CS3 via resistor or solder jumper	analogue output		A6	
OUT_CS2	Output signal current sensor phase L2/V	analogue output	$+I_{max}$ … $-I_{max}$ -> 0V … 5V see chapter 2.4		X22:15
OUT_CS2_S	Connectable and scalable to OUT_CS2 via resistor or solder jumper	analogue output		A4	
REF_CS2	Reference voltage current sensor phase L2/V	analogue output	V = 2.5V see chapter 2.4		X22:16
REF_CS2_S	Connectable and scalable to REF_CS2 via resistor or solder jumper	analogue output		A7	
OUT_CS1	Output signal current sensor phase L1/U	analogue output	$+I_{max}$ … $-I_{max}$ -> 0V … 5V see chapter 2.4		X22:17
OUT_CS1_S	Connectable and scalable to OUT_CS1 via resistor or solder jumper	analogue output		A5	
REF_CS1	Reference voltage current sensor phase L1/U	analogue output	V = 2.5V see chapter 2.4		X22:18
REF_CS1_S	Connectable and scalable to REF_CS1 via resistor or solder jumper	analogue output		A8	
5V	5V supply voltage (required for current sensors) connectable to optional linear voltage regulator on adapter board or connectable to supply voltage provided by controller board; see chapter 2.2	power	V = 5V ±5% I_{max} = 100mA	5V0	X22:20
3V3	3.3V supply circuit connectable to optional linear voltage regulator on adapter board or connectable to supply voltage provided by controller board; see chapter 2.2	power	V = 3.3V ±5% I_{max} = 20mA	3V3	
Vcc_GD_S	Connectable and scalable to Vcc_GD via resistor; see chapter 2.2	analogue output		A12	
CAN_H	CAN bus High signal	Output/input	see chapter 2.8		
CAN_L	CAN bus Low signal	Output/input	see chapter 2.8		
CAN_RX	Receive signal CAN transceiver	Output/input	see chapter 2.8	A14	
CAN_TX	Transceive signal CAN transceiver	Output/input	see chapter 2.8	A15	
LIN4	Reserved; do not connect or route these pins!			4	
IOREF	Reference voltage option from controller; see chapter 2.2			IOREF	
NC	Reserved; do not connect or route these pins!			NC	

Figure 4: SKAI3 LV Evaluation Kit Version

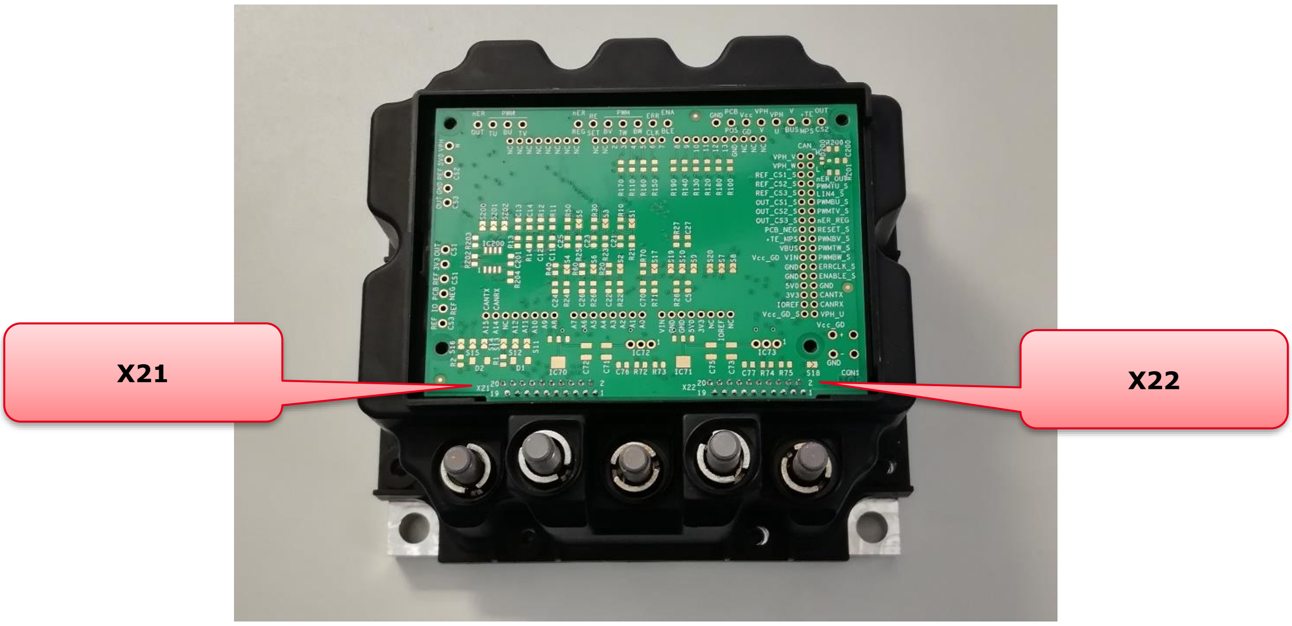
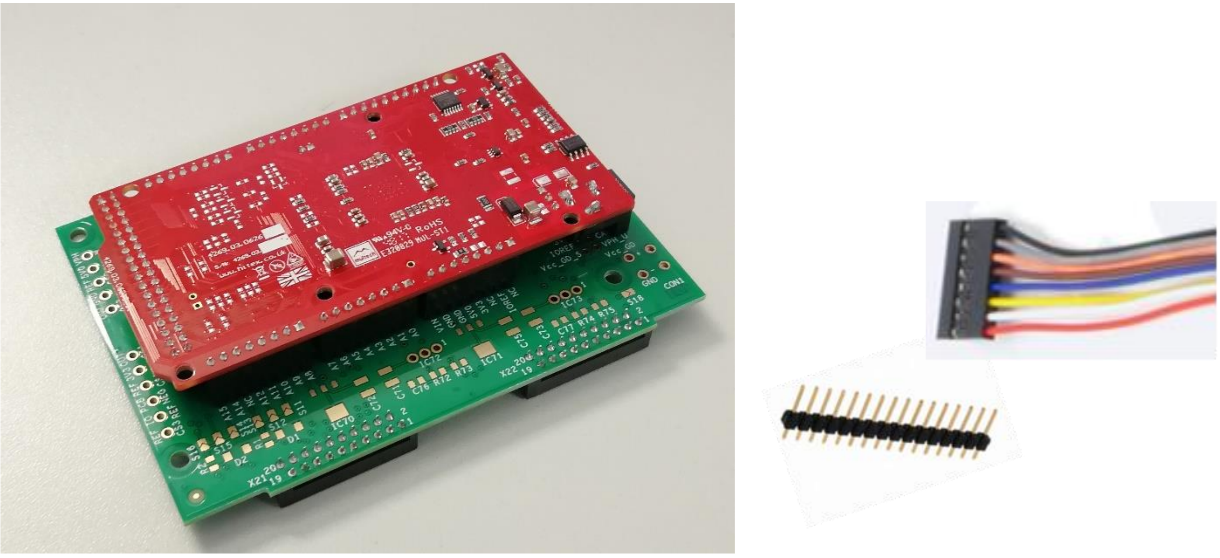


Figure 5: Examples: controller assembly on eval-kit adapter board (left) / suitable connectors for cable connections (right)



2.2 Power supplies

SKAI3 LV uses separate circuits that are electrically insulated between driver/power stage circuitry and current/temperature measurement circuitry. SKAI3 LV-eval-kit adapter board uses a simplified power supply concept, which is sufficient for evaluation purposes.

- Respective references (GND) of power supplies for driver, current/temperature measurement are directly connected on the SKAI3 LV-eval-kit adapter board (see Figure 6).
- Connection from DC- to GND can be optionally realized by shorting the solder-jumper S18.
- SKAI3 LV can either be supplied over CON1 on eval-kit adapter board or from controller board by shortening the solder-jumper S17

- 5V supply 5V0 is required for current sensor supply. It can be generated from Vcc_GD by populating one of the linear regulator IC71 / IC73 or connecting external 5V supply from controller board
- 3.3V supply 3V3 can either be generated from Vcc_GD by populating one of the linear regulators IC70 / IC72 or connecting external 3.3V supply from controller board
- If additional voltage level is required it can be externally connected from controller board via

Connector CON1 can be populated with Phoenix MSTBA 2,5/ 2-G-5,08 (manufacturer ID: 1757242) connector or other compatible connector with 5.08 mm pitch.

Figure 6: Power supply concept

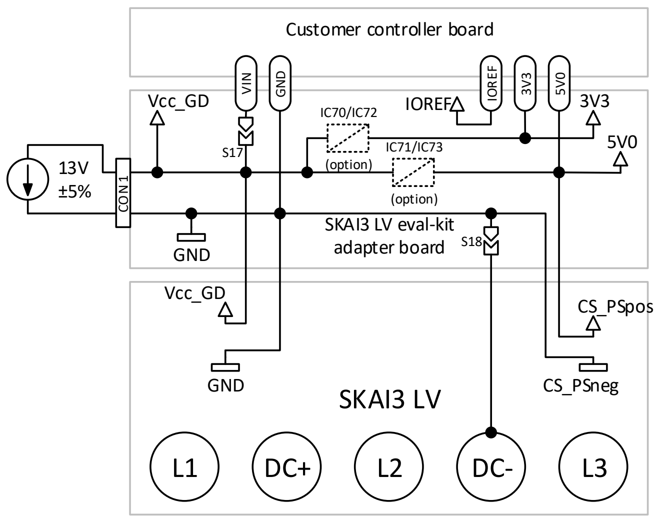
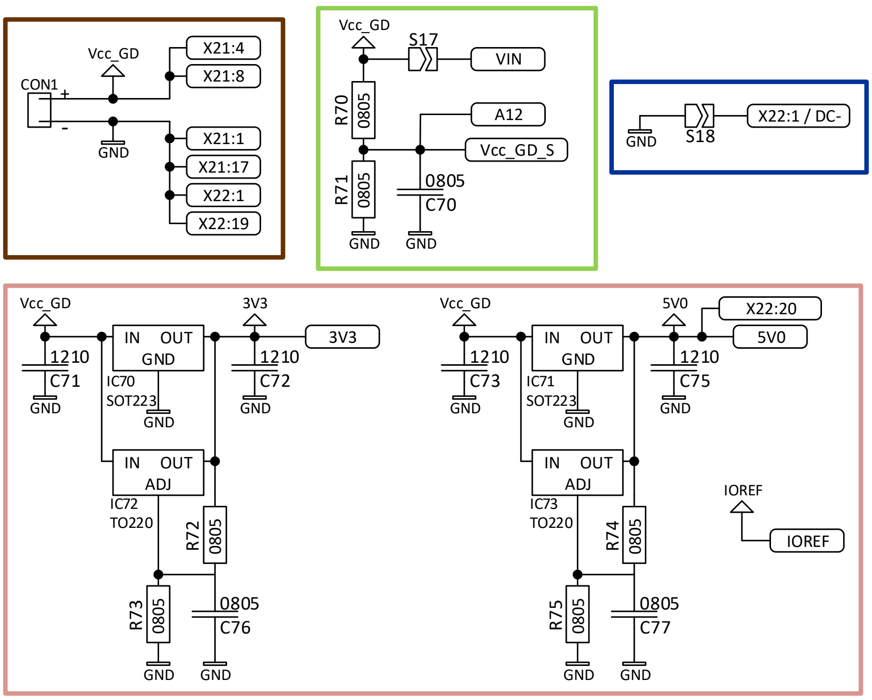


Figure 7: schematic of power supply assembly options

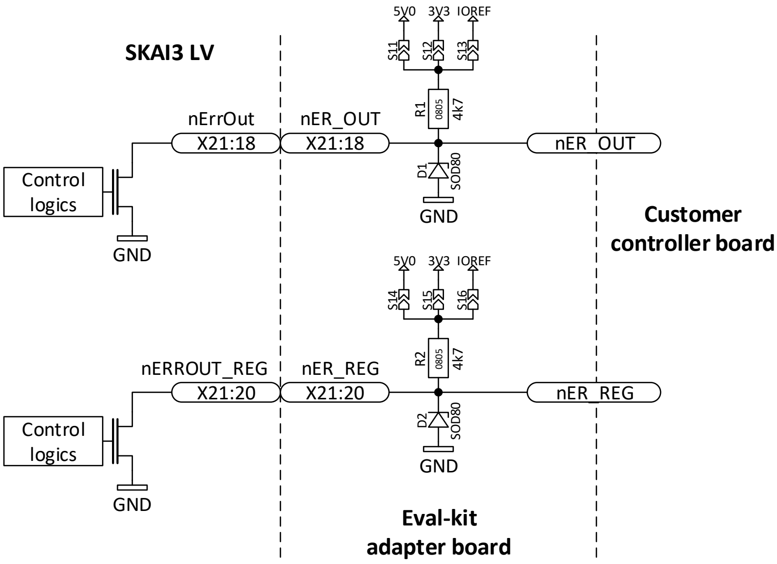


2.3 Digital- and PWM- signals

2.3.1 Digital outputs

The signals "nERROUT" and "nERROUT_REG" are open drain outputs and shall be connected via pull up resistor in a range between 2 kΩ and 5.5 kΩ to microcontroller power supply (see Figure 8).

Figure 8: Digital output circuit and voltage limitation (nER_OUT, nER_REG)

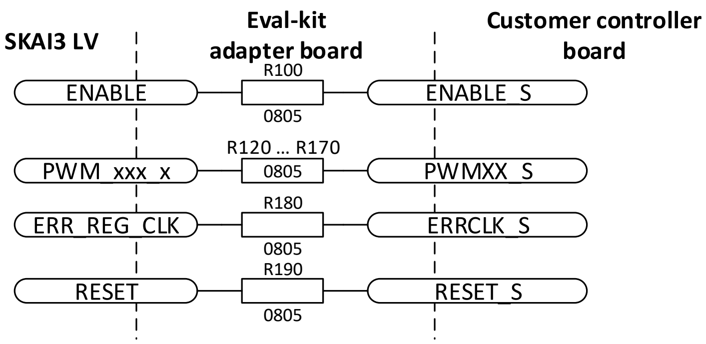


The positions of the resistors R100 to R190 can be found on the eval-kit adapter board in the Figure 2 **(turkey)**.

2.3.2 Digital inputs

Please solder proper series resistors 0805 on positions R100, R120, R130, R140, R150, R160, R170, R180 and R190 according to controller specification before operating the SKAI3 LV eval-kit. 100Ω resistors can be used as default.

Figure 9: Digital input series resistor circuit diagram



The positions of the resistors R100 to R190 can be found on the eval-kit adapter board in the Figure 2 **(green)**.

2.4 Current sense circuits

SKAI3 LV provides three current sensors for phase current measurement. Current sensor scales measured value as follows: $V_{OUT} = (V_{5V}/5V) \cdot (V_O - G \cdot I)$; $V_O = 2.5V$; $G = 2.22\,mV/A$. For more details see Technical Explanation SKAI3 LV [3].

Note: The current sensors use the 5V supply from the adapter board or linear regulator populated on eval-kit adapter board as described in chapter 2.2. Sensing accuracy depends directly on the accuracy of supply voltage connected to the adapter board by the user.

Figure 10: Current sensor circuit with signal conditioning (Sx = solder jumper) for L1/U

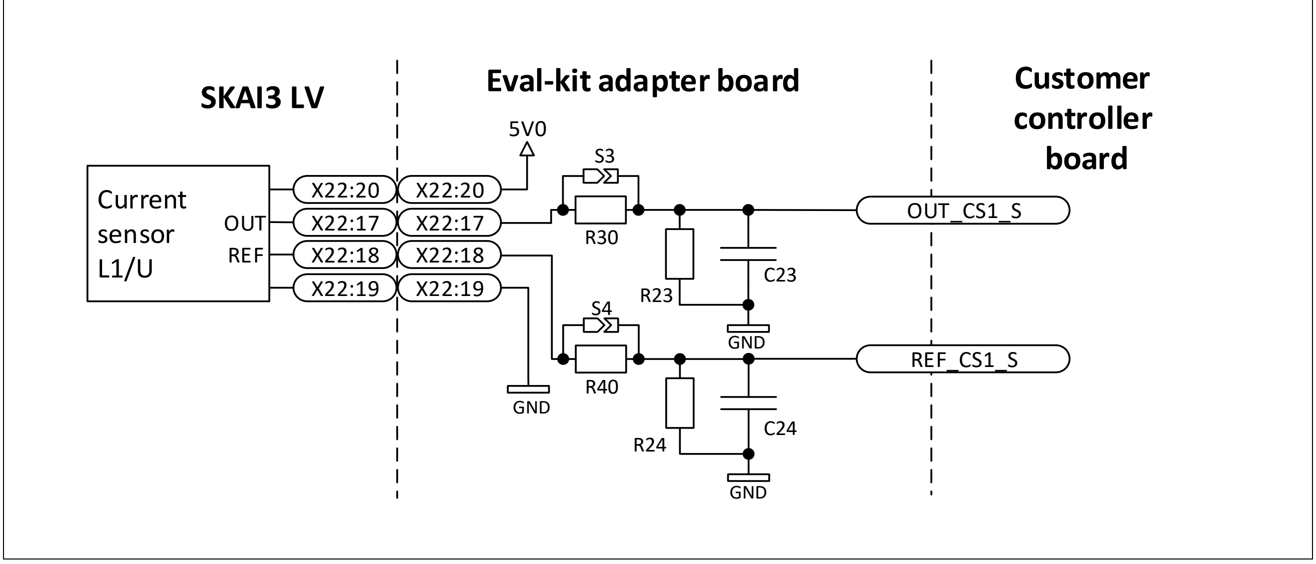


Figure 11: Current sensor circuit with signal conditioning (Sx = solder jumper) for L2/V

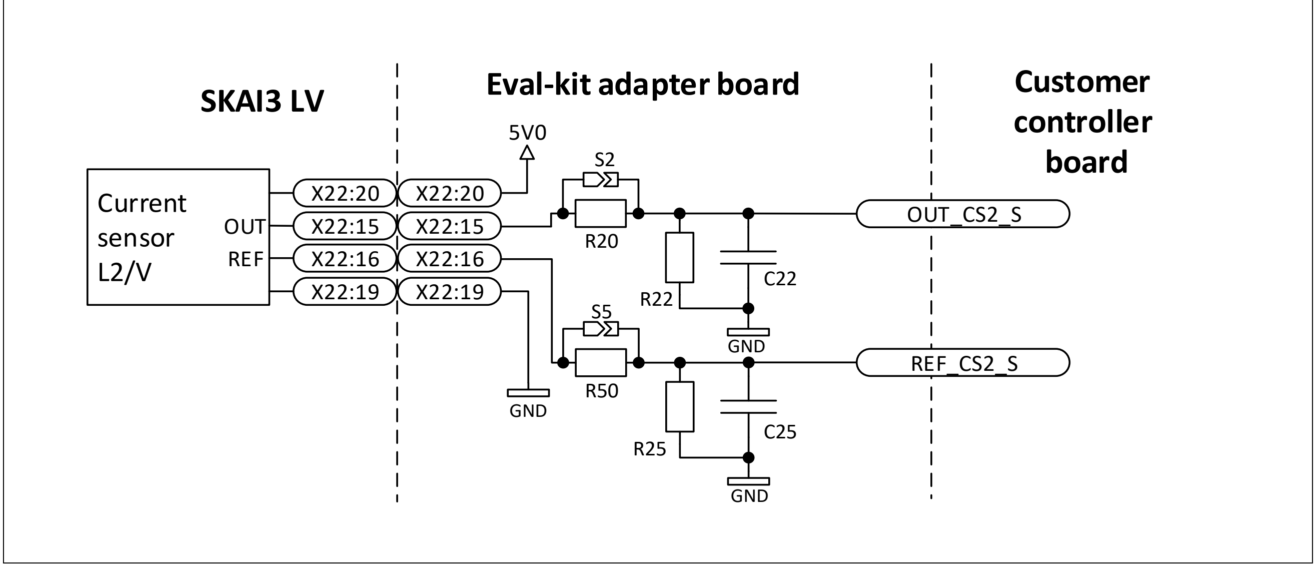
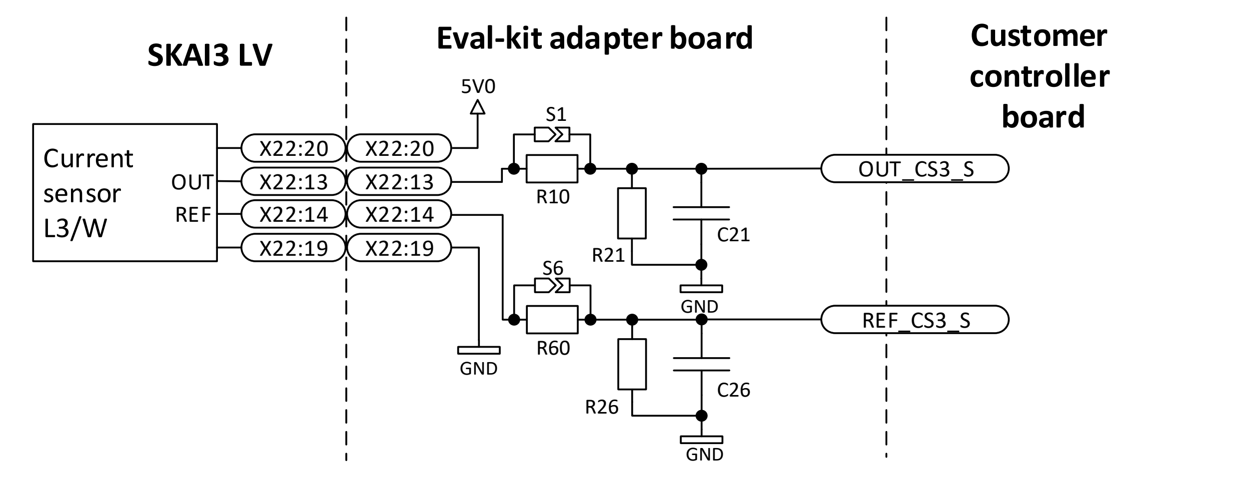


Figure 12: Current sensor circuit with signal conditioning (Sx = solder jumper) for L3/W



The SKAI3 LV eval-kit adapter board provides placement area for signal conditioning purposes as shown in Figure 10, Figure 11 and Figure 11 to meet the individual requirements on signal quality and range. Component shape type is 0805. Table 2 lists some example values. Please insert the components before operating the SKAI3 LV eval-kit.

Table 2: Examples of component selection for current sensor signal conditioning circuit

Voltage scaling	Filter frequency $f_{0.PT1}$	Solder jumper S1/S2/S3/ S4/S5/S6	R10/R20/R30/ R40/R50/R60	R21/R22/R23/ R24/R25/R26	C21/C22/C23/ C24/C25/C26
0…5V	No filter	Set	Not installed	Not installed	Not installed
0…5V	19.4 kHz	Not set	1.0 kΩ	Not installed	8.2 nF
0…3.3V	20.2 kHz	Not set	5.6 kΩ	10.0 kΩ	2.2 nF
0…3.0V	21.8 kHz	Not set	6.8 kΩ	10.0 kΩ	1.8 nF

The positions of the resistors R10, R20, R30, R40, R50, R60, R21, R22, R23, R24, R25 and R26 can be found on the eval-kit adapter board in the Figure 2 **(red)**.

The positions of the capacitors C21, C22, C23, C24, C25 and C26 can be found on the eval-kit adapter board in the Figure 2 **(red)**.

The positions of the solder jumper S1, S2, S3, S4, S5 and S6 can be found on the eval-kit adapter board in the Figure 2 **(red)**.

2.5 Voltage sense circuits

SKAI3 LV provides four voltage sense circuits for DC link- and three phase-voltages. Initially, the voltage sense circuits scale the DC link voltage 0…V_{max} to 0…5V; the diagram below shows the respective voltage divider. Please insert 0805 shape components on eval-kit adapter board if other scaling factor or additional filtering is required.

Figure 13: Voltage sense circuit SKAI3 LV with signal conditioning

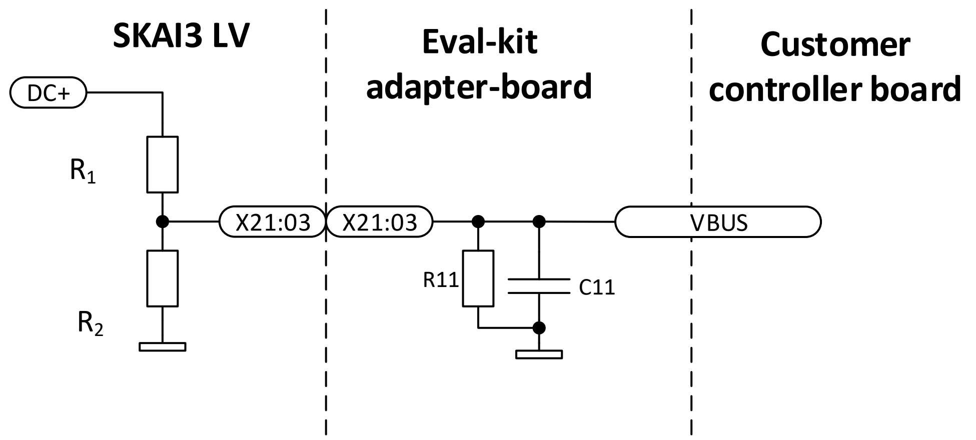


Figure 14: Voltage sense circuit SKAI3 LV with signal conditioning

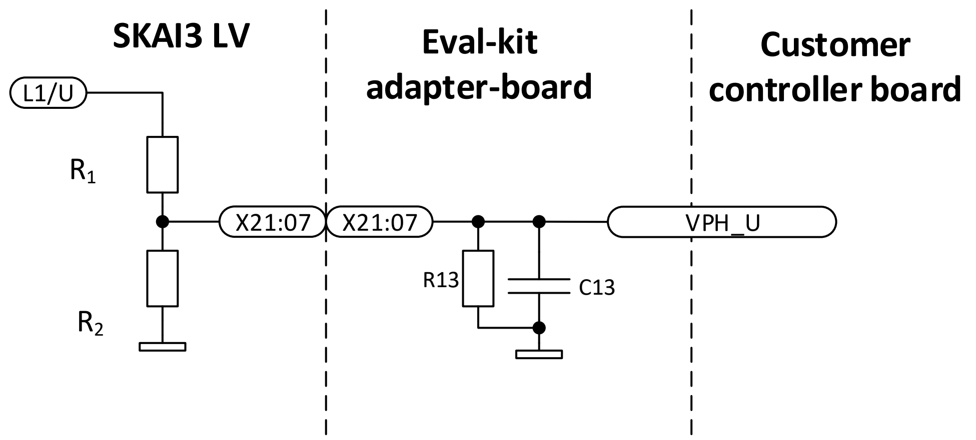


Figure 15: Voltage sense circuit SKAI3 LV with signal conditioning

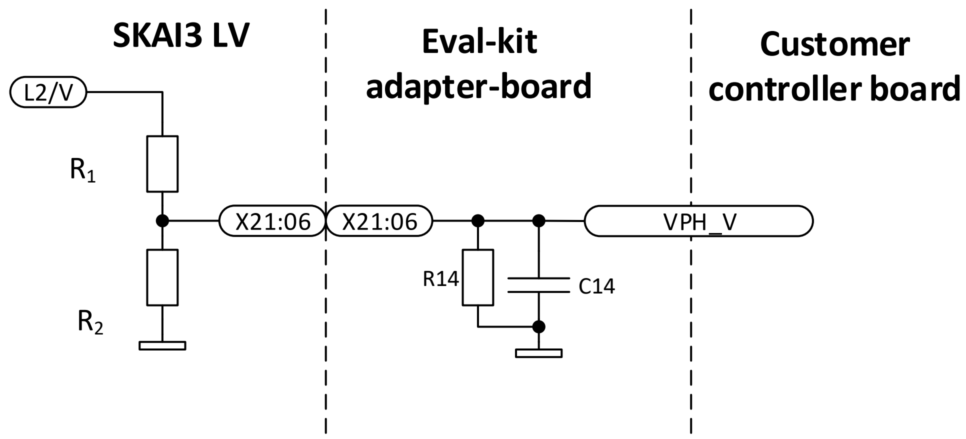


Figure 16: Voltage sense circuit SKAI3 LV with signal conditioning

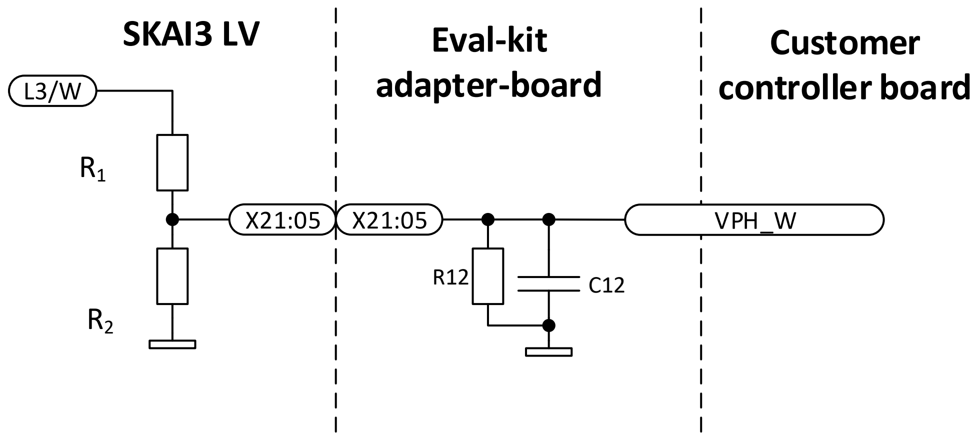


Table 3: Voltage divider resistor values

SKAI3 LV voltage class	R_1	R_2
SKAI XX A3 MD10-X	225 kΩ ± 1%	13.0 kΩ ± 1%
SKAI XX A3 MD15-X	330 kΩ ± 1%	13.0 kΩ ± 1%

Following Table lists some example values.

Table 4: Examples of component selection for voltage-sense signal conditioning circuit

Vphase_X_scal, Vbus	Filter frequency $f_{0.PT1}$	R11/R12/ R13/R14	C11/C12/ C13/C14
0...5V	1.1 kHz	Not installed	12 nF
0...3.3V	1.1 kHz	24 kΩ	18 nF
0...3.0V	1.0 kHz	18 kΩ	22 nF

The positions of the resistors R11, R12, R13 and R14 as well as the capacitors C11, C12, C13 and C14 can be found on the eval-kit adapter board in the Figure 2 **(blue)**.

2.6 DCB temperature circuit

Power-module temperature sensor represents the temperature of DCB (direct copper bonded ceramic board) and is an indicator of the actual power MOSFET chip temperature.
Note: Details on the DCB-temperature sensor are described in the Technical Explanations document [3].

Insert the 0805 chip resistor R28 to adjust the required temperature-/voltage- scaling, for example 1.0 kΩ. Set the solder jumper according to the connected ADC pin specification to connect the voltage divider to either 5V or 3.3V supply.

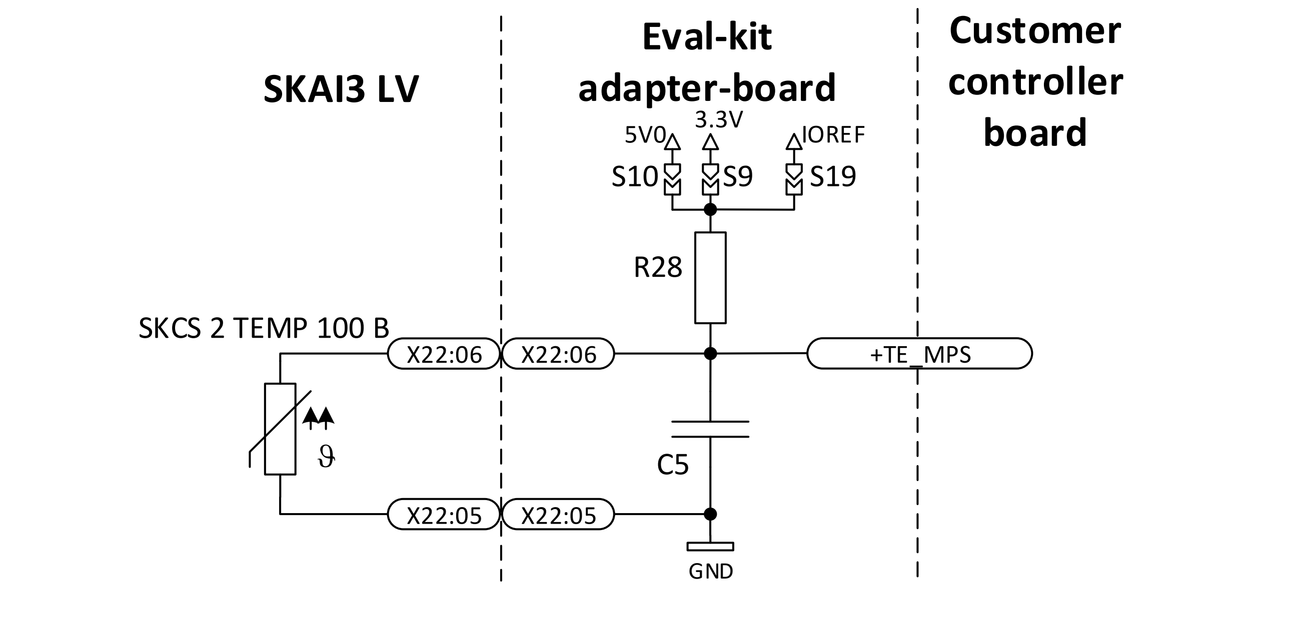
Insert the 0805 capacitor C5 if analogue filtering is required, for example 1.0 µF.

The positions of the resistors R28 can be found on the eval-kit adapter board in the Figure 2 **(purple)**.

The positions of the capacitors C5 can be found on the eval-kit adapter board in the Figure 2 **(purple)**.

The positions of the solder jumpers S9, S10 and S19 can be found on the eval-kit adapter board in the Figure 2 **(purple)**.

Figure 17: DCB (Power module) temperature sensing circuit



2.7 PCB temperature circuit

PCB temperature sensor is placed on a passive area of the driver PCB of SKAI3 LV. It represents average air-temperature of the electronics ambient temperature.
Note: Details on the PCB temperature sensor are described in the Technical Explanations document [3].

Insert the 0805 chip resistor R27 to adjust the required temperature-/voltage- scaling, for example 1.0 kΩ. Set the solder jumper according to the connected ADC pin specification to connect the voltage divider to either 5V or 3.3V supply.

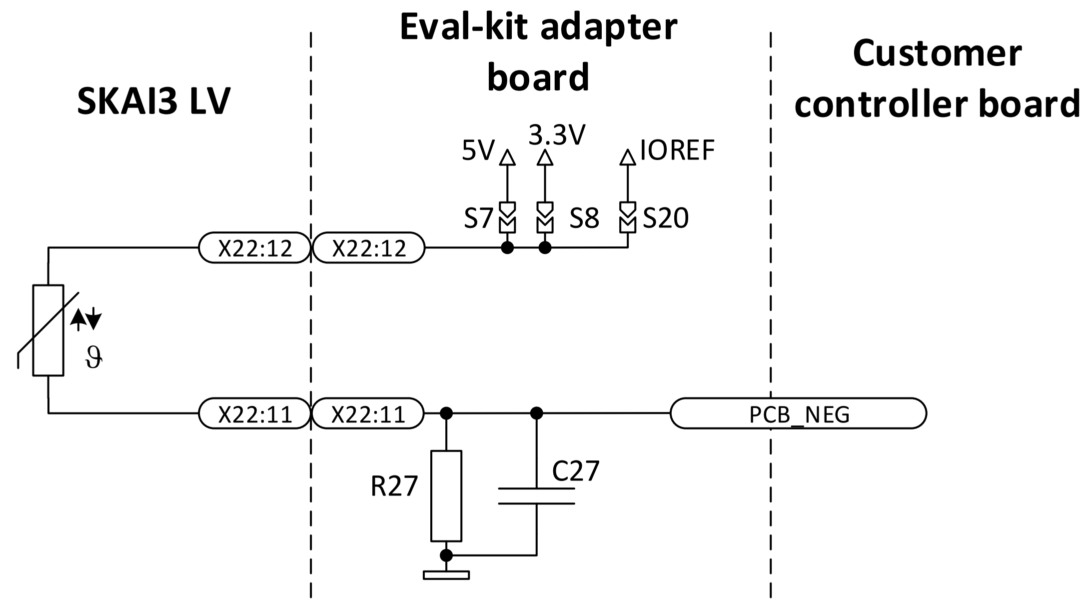
Insert the 0805 capacitor C27 if analogue filtering is required, for example 1.0 µF.

The positions of the resistors R27 can be found on the eval-kit adapter board in the Figure 2 **(orange)**.

The positions of the capacitors C27 can be found on the eval-kit adapter board in the Figure 2 **(orange)**.

The positions of the solder jumper S7 and S8 can be found on the eval-kit adapter board in the Figure 2 **(orange)**.

Figure 18: PCB temperature sensing circuit

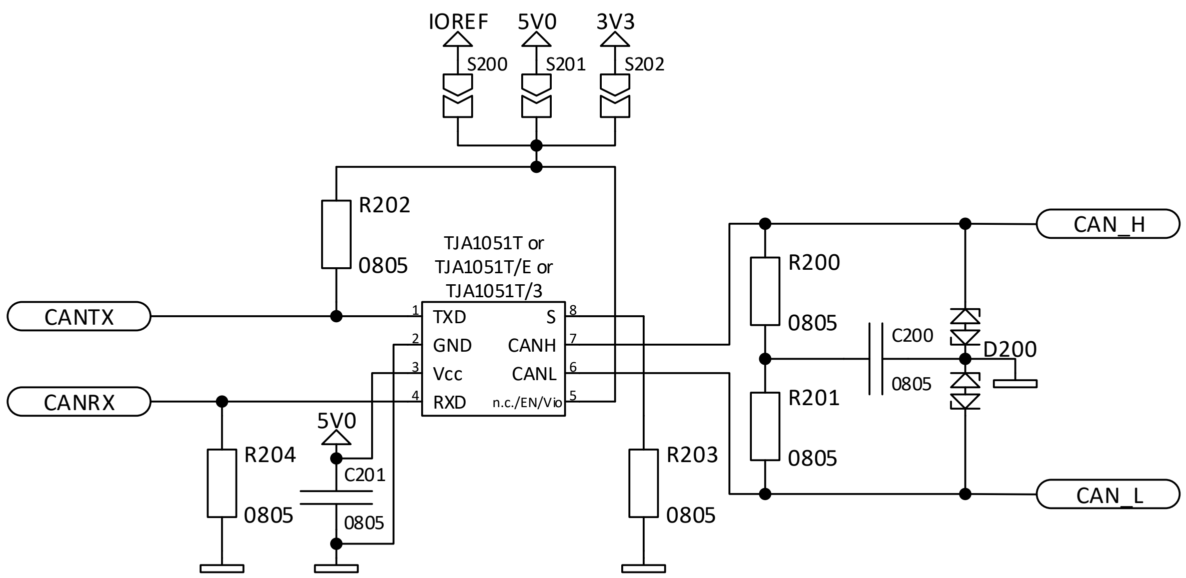


2.8 CAN transceiver circuit

Optional CAN transceiver NXP TJA1051T, TJA1051T/E or TJA1051T/3 can be populated on the eval-kit adapter board if can communication is required. The positions of CAN transceiver and its related circuit components can be found in the Figure 2 **(light brown)**.

CAN transceiver and component selection please refer to CAN specification and installed transceiver IC datasheet.

Figure 19: CAN transceiver circuit



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Symbols and terms

Letter Symbol	Term
SKAI3 LV	SEMIKRON Advanced Low Volt Inverter 3-rd generation
DCB	Direct bonded copper

A detailed explanation of the terms and symbols can be found in the "Application Manual Power Semiconductors" [2]

References

[1] www.SEMIKRON.com
[2] A. Wintrich, U. Nicolai, W. Tursky, T. Reimann, "Application Manual Power Semiconductors", 2nd edition, ISLE Verlag 2015, ISBN 978-3-938843-83-3
[3] Technical Explanation SKAI3 LV
[4] SKAI3_LV_Eval Kit_Adapter Board V1.1 (outdated document version describing previous adapter board layout)

IMPORTANT INFORMATION AND WARNINGS

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